OASIS-1 Exploratory Data Analysis: Task 1 Report

Dataset: OASIS-1 cross-sectional (Kaggle: oasis-1-shinohara)

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1. Objective

This report presents an exploratory data analysis (EDA) of the OASIS-1 cross-sectional MRI dataset. The goal is to understand the demographic and clinical data, the structure and quality of the brain scans, and the class labels (Clinical Dementia Rating, CDR), and to identify problems that must be handled before building a dementia classifier. The analysis has two parts: (1) the demographic and clinical CSV, and (2) the neuroimaging files (NIfTI, .nii).

2. Related Work

2.1 Comparison of Studies

Eleven papers from 2023 to 2026 on MRI-based Alzheimer's disease (AD) classification are compared below by dataset, task, method and headline metrics: nine report experiments and two are reviews.

Title, authors, year	Dataset	Application area	Method / model	Headline metrics
Research on classification method of multimodal magnetic resonance images of Alzheimer's disease based on generalized convolutional neural networks. Qin et al., 2023, J Biomed Eng	ADNI; 91 subjects (44 AD, 47 NC); sMRI + rs-fMRI	AD vs NC classification	gCNN: 3D HA-ResUNet (sMRI) + U-GCN (fMRI brain network); feature fusion, binary PSO feature selection, ML classifier	ACC 88.89%, SEN 100%, SPE 77.78%, G-mean 88.19% (best single modality, U-GCN: ACC 83.33%)
s2MRI-ADNet: an interpretable deep learning framework integrating Euclidean-graph representations of Alzheimer's	ADNI, AIBL, EDSD, OASIS; 3,377 subjects; sMRI only	NC vs AD classification, within and across databases	Dual channel: GMV maps with CNN-Transformer + regional radiomics similarity network (R2SN) with GraphSAGE and	Intra-database ACC 92.1%, AUC 0.962; inter-database ACC 91.4%, AUC 0.938

Title, authors, year	Dataset	Application area	Method / model	Headline metrics
disease solely from structural MRI. Song et al., 2024, MAGMA			MLP	
A Graph Convolutional Network Based on Univariate Neurodegeneration Biomarker for Alzheimer's Disease Diagnosis. Qu et al., 2023, IEEE JTEHM	ADNI, Aβ+ subjects; sMRI cortical thickness (FreeSurfer, left hemisphere, 34 ROIs)	AD vs CU and AD vs MCI classification	GCN on a population graph (age, gender, APOE edges) with univariate neurodegeneration biomarker (UNB) node features and an attention module	Validation ACC 93.90% (AD vs CU), 82.05% (AD vs MCI)
A Hybrid Deep Learning Framework to Predict Alzheimer's Disease Progression Using Generative Adversarial Networks and Deep Convolutional Neural Networks. SinhaRoy & Sen, 2024, Arab J Sci Eng	ADNI-derived 4-class set of 2D sMRI slices; 6,400 images	4-stage AD classification	DCGAN (synthetic images) + SRGAN (super-resolution) + compact 2D CNN	ACC 99.7% (AD vs NL AUC 1.000)
An Alzheimer's Disease classification network based on MRI utilizing diffusion maps for multi-scale feature fusion in graph convolution. Yang	ADNI; 338 AD, 422 MCI, 532 CN; sMRI ROI features (116 AAL regions)	AD / CN / MCI pairwise and 3-class classification	Population graph; diffusion maps at multiple neighbour counts; metric-learned adjacency; self-attention fusion	ACC: AD-CN 0.887, AD-MCI 0.705, CN-MCI 0.825, 3-class 0.671

Title, authors, year	Dataset	Application area	Method / model	Headline metrics
et al., 2024, Math Biosci Eng				
Graph-Based Multimodal Fusion Framework with Correlation-Aware Learning for Alzheimer's Disease Prediction. Li et al., 2026, Big Data Min Anal	ADNI; 417 subjects (103 CN, 103 EMCI, 105 LMCI, 106 AD); sMRI + SNP genetics	CN vs AD, CN vs MCI, MCI vs AD classification	Brain-region graph + gene-ontology graph; GCN with pooling; cross-attention + correlation-aware learning (CAL) module	CN vs AD: ACC about 0.82, AUC 0.82; CN vs MCI: ACC 0.773; MCI vs AD: ACC 0.747
Longitudinal Graph Neural Networks for Prognostic Modeling of Alzheimer's Disease Progression. Ashimgaliyev et al., 2026, AAIML	ADNI; about 600 MCI subjects; longitudinal sMRI + FDG-PET / amyloid-PET features	Predicting MCI-to-AD conversion within 5 years	Patch-based 3D CNN embeddings as nodes in a spatiotemporal graph (82 ROIs); GAT (spatial) + GRU with temporal attention	AUC 0.89, C-index 0.84, BAC 86%, SEN 85%, SPE 87%
A multilayered framework for diagnosis and classification of Alzheimer's disease using transfer learned Alexnet and LSTM. Goyal, Rani & Singh, 2024, Neural Comput Appl	ADNI T1 MRI; 200 subjects (60 AD, 73 MCI, 67 CN); 5,980 2D slices	3-class (AD / MCI / CN) and pairwise classification	Pre-trained AlexNet features, then LSTM, then transfer-learned dense layers; GAN augmentation of training data	With GAN: ACC 96.83% (3-class), 98.13% (AD-CN), 99.38% (AD-MCI), 99.37% (CN-MCI)
Machine Learning and Deep Learning for Alzheimer's Disease Diagnosis:	Survey, no dataset of its own; covers ADNI, OASIS-3, AIBL	Review of ML/DL for AD diagnosis (MRI, PET, multimodal)	Narrative review organised by traditional ML, CNN, multimodal fusion,	None (no experiments). Concludes multimodal usually beats

Title, authors, year	Dataset	Application area	Method / model	Headline metrics
A Survey of MRI, PET, Multimodal Fusion, Transformers, Graph Neural Networks, and Explainable AI. Ahmad, Tarannum & Uddin, 2026, Am J Adv Technol Eng Solut			transformers, GNN and explainable AI	single-modality; transformers and GNNs promising but not proven superior
Graph neural networks in Alzheimer's disease diagnosis: a review of unimodal and multimodal advances. Ali, Piana, Pardini & Garbarino, 2025, Front Neurosci	73 reviewed studies; mostly ADNI, also TADPOLE, OASIS, UK Biobank, ABIDE	Review of GNN-based AD diagnosis (NC/AD, NC/MCI, MCI/AD, progression)	Review organised by GNN pipeline: graph construction, convolution, pooling, prediction, explainability	None of its own. Best reported 99.3% NC/AD (TADPOLE, multimodal); multimodal studies had higher median accuracy than unimodal
Development of framework by combining CNN with KNN to detect Alzheimer's disease using MRI images. Lanjewar, Parab & Shaikh, 2023, Multimed Tools Appl	Kaggle 4-class Alzheimer's MRI set; 6,400 images (ND 3,200, VMD 2,240, MID 896, MOD 64)	4-stage AD classification	4-block CNN as feature extractor feeding a KNN (k = 3); compared with ResNet50, VGG16, MobileNetV2	Mean ACC 99.58% (K = 20), F1 99.43%, MCC 99.31%

The highest accuracies (96.8% to 99.7%) come from the three studies that classify 2D slice or image sets (SinhaRoy & Sen, Goyal et al., Lanjewar et al.). The graph and multi-database studies report about 82% to 94% accuracy for AD vs CN and 67% to 83% for the harder MCI pairs and 3-class tasks. The table does not say whether the slice-based splits were made by subject, so these figures are context, not targets for this project; the closest comparison is the structural-MRI work with AUC 0.82 to 0.96 for AD vs CN (Li et al., Song et al.), set against our age-restricted baseline in Section 10.2.

2.2 Research Gap

Four gaps recur across the six graph-based studies (Qin et al., Yang et al., Qu et al., Song et al., Li et al., Ashimgaliyev et al.):

1. None builds a graph from the pixels of each MRI image (superpixels or patches). Graphs come from atlas regions, subject similarity or connectivity.
2. Five of the six use ADNI only. The one study that includes OASIS (Song et al.) reports pooled results, and its OASIS cohort shows a large age gap between normal and AD groups (NC 67.1 vs AD 75.0 years, $p < 0.001$) with no age control described.
3. Leakage controls are often unstated (subject-level splitting, graph built before splitting), and several headline gains rest on single random splits of small samples. For Qin et al. (91 subjects, about 18 in the test set) one subject equals 5.56 accuracy points.
4. Ablations of the graph itself (construction method, edge weights) are rare.

Study	Gap the study leaves	Relation to our work
Qin et al., 2023	The graph comes from fMRI time series, not from the image itself; very small sample	Shows an attention-CNN branch and a graph network can be combined; our graph comes from the structural image
Yang et al., 2024	Graph over patients built on pre-extracted region vectors, so images never become graphs; accuracy only	GCN with self-attention fusion on MRI, the nearest to our GCN + attention; we build a graph inside each image instead
Qu et al., 2023	Patient-level graph; age is used as an edge but not controlled	Attention on node features and phenotype edges; we handle age by using subjects aged 60 and over and splitting by subject
Song et al., 2024	Atlas-based graphs, not image-derived; its OASIS cohort differs in age (NC 67.1 vs AD 75.0 years, $p < 0.001$) and no age control is described	Closest data match (OASIS, 91×109×91 at 2 mm) and our strongest comparison candidate, but it reports no OASIS-only result
Li et al., 2026	Needs genetics; region graph computed across samples; no OASIS data	Reference design for cross-attention and for ablation plus 5-fold reporting like our Task 3
Ashimgaliyev et al., 2026	Needs longitudinal PET; uses region patches, not a graph over a whole slice	CNN embeddings as graph nodes is the idea behind our CNN + GCN; its split, confidence intervals and significance tests are the protocol we must follow

Our proposed contribution is a GCN + CNN + attention model that builds a per-image graph (superpixel or patch nodes) from T1 slices. It is evaluated on OASIS-1 with subject-wise splits, an age-controlled subset of subjects aged 60 and over (Section 10.1), 5-fold cross-validation, an ablation of the graph and a significance test against the best CNN baseline. Each figure above is quoted under its own paper's cohort and split, so the results are not comparable with each other or with our OASIS results.

3. Dataset Description

The CSV contains 436 rows (one per MRI session) and 12 columns, covering 416 unique subjects. Twenty subjects have a second session (MR2), so the data holds repeated measures and must be split by subject. Ages range from 18 to 96 (mean 51.4). There are 268 female and 168 male sessions. All subjects are right-handed, so the Hand column is constant and carries no information.

Column	Meaning
ID	Session ID (e.g. OAS1_0001_MR1)
M/F	Sex
Hand	Handedness (all R)
Age	Age at scan
Educ	Education code, 1 to 5
SES	Socioeconomic status, 1 to 5 (1 = highest)
MMSE	Mini-Mental State Examination, 0 to 30
CDR	Clinical Dementia Rating (0, 0.5, 1, 2)
eTIV	Estimated total intracranial volume (mm^3)
nWBV	Normalised whole-brain volume
ASF	Atlas scaling factor
Delay	Days between MRI and clinical assessment (repeat-scan subjects)

The image set has 436 NIfTI files, one per session. A one-to-one match with the CSV was confirmed: no CSV row lacks an image and no image lacks a row. The filenames (for example mpr_n4_anon_sbj_111_normalised) suggest N4 bias-corrected, atlas-registered, intensity-normalised images. Of the 436 files, 434 follow the standard naming pattern and 2 follow a different pipeline variant (fdcanon).

4. Data Quality

4.1 Missing values and duplicates

Column	Missing	%

Column	Missing	%
Delay	416	95.4
SES	220	50.5
Educ	201	46.1
MMSE	201	46.1

About 19.8% of all cells in the original 12 columns are missing. There are no duplicate session IDs.

The missingness is structural rather than random. MMSE, Educ and SES are missing for almost all subjects under 30 (100%) and become complete from age 60, because younger subjects were not clinically assessed. Delay is present only for the 20 repeat-scan subjects.

Age group	MMSE missing	SES missing
under 30	1.00	1.00
30 to 39	0.88	0.88
40 to 49	0.58	0.61
50 to 59	0.33	0.33
60 to 69	0.00	0.15
70 to 79	0.00	0.11
80 and over	0.00	0.04

Importantly, CDR is not missing for any age group. In this copy of the data CDR appears to have been filled in as 0 for the young subjects, who were never clinically assessed. This affects how the labels should be interpreted (Section 10).

4.2 Repeated sessions

There are 416 MR1 sessions and 20 MR2 sessions. Repeat scans of the same person are highly similar, so a random scan-level split would leak information between training and test sets. Splits must be made at the subject level.

5. Demographic Distributions

- Age is bimodal, with one peak at about 20 to 25 years and another at about 70 to 80. The mean of 51 describes very few subjects.
- The sample is 61% female and 39% male.
- MMSE has a strong ceiling effect: about 120 of 235 scores equal 30, and the distribution is left-skewed with a minimum of 14.
- eTIV is roughly bell-shaped (mean 1482 mm^3). nWBV (0.644 to 0.893) has a left tail and a peak near 0.84, reflecting the mix of young and old brains. ASF mirrors eTIV.
- Education and SES codes are spread across their ranges. SES level 5 has very few subjects.

6. Class Distribution

CDR	Class	Count	%	Inverse-frequency weight
0	Nondemented	336	77.1	0.32
0.5	Very mild	70	16.1	1.56
1	Mild	28	6.4	3.89
2	Moderate	2	0.5	54.50

The four-class problem is severely imbalanced (168:1 between the largest and smallest class). The Moderate class has only 2 scans and cannot be split across training, validation and test sets, and a class weight of 54.5 would let two images dominate the loss. A binary label (CDR 0 versus CDR greater than 0: 336 versus 100, a 3.4:1 ratio) or a three-class label that merges Mild and Moderate is more realistic. A model that always predicts Nondemented would reach 77% accuracy, so accuracy alone is a misleading metric.

7. Clinical Variables Versus CDR

7.1 Group differences

- nWBV falls steadily with CDR (median about 0.83, 0.73, 0.70 and 0.685 for CDR 0, 0.5, 1 and 2).
- MMSE falls with CDR. Both Moderate subjects have MMSE 15.
- Age rises with CDR, but the Nondemented class spans 18 to 94 years while all other classes start at about 62.
- eTIV and ASF are flat across classes and carry no disease signal.
- Education shows a weak downward trend with CDR.

Sex by CDR (counts of Nondemented / Very mild / Mild / Moderate): females 209 / 39 / 19 / 1, males 127 / 31 / 9 / 1. About 22% of females and 24% of males have CDR above 0.

7.2 Statistical tests

Kruskal-Wallis tests (rank-based, suitable for skewed and ordinal variables) compared the four CDR groups. A chi-square test compared sex with the binary label.

Variable	Test statistic	p-value
Age	H = 122.4	2.4e-26
MMSE	H = 117.7	2.4e-25
nWBV	H = 159.2	2.7e-34
Education	H = 15.4	0.0015
eTIV	H = 0.2	0.98

Variable	Test statistic	p-value
Sex vs demented	chi-square = 0.21	0.65

Age, MMSE and nWBV differ strongly between groups; eTIV and sex do not. Two cautions apply. The age result is largely driven by the young Nondemented subjects, and the education result is probably age-related because older cohorts had less schooling. The Moderate group (n = 2) adds almost no information.

7.3 Relationships between variables

NWBV declines roughly linearly with age. At the same age, demented subjects sit below the Nondemented trend, but among subjects over 70 the classes overlap considerably. MMSE for Nondemented subjects hugs the ceiling of 30.

Spearman correlations (rank-based):

- Age and nWBV: -0.87 (brain volume mostly tracks age)
- CDR and MMSE: -0.71
- CDR and nWBV: -0.60
- CDR and Age: 0.53
- eTIV and ASF: -0.99 (redundant; only one should be used in a model)
- Education and SES: -0.74 (partly because SES is coded 1 = highest)

The matrix mixes two populations: Age, CDR and nWBV cover all subjects, while MMSE, Education and SES exist mainly for subjects aged 60 and over.

8. Neuroimaging Analysis

8.1 Geometry and format

All 436 volumes are identical in format: shape 91 x 109 x 91, 2 mm isotropic voxels, LAS orientation, float32, and 3.61 MB each (this equals 91 x 109 x 91 x 4 bytes plus the header). The grid matches a standard 2 mm atlas space. No resampling or cropping is required, and image resolution cannot be confounded with the label.

8.2 Visual inspection

Mid-slice views per class show larger ventricles and wider sulci in the Very mild and Mild examples than in young Nondemented examples. However, older Nondemented subjects (for example age 82) also show enlarged ventricles, so visual atrophy overlaps with normal ageing. One Moderate scan (OAS1_0351_MR1) shows a different anatomical level (skull base and sinuses rather than mid-brain), indicating a misregistered or shifted volume, not pathology.

A class-similarity analysis (correlation of each class's mean slice after removing the grand mean) found the Very mild and Mild classes most similar (r = 0.40), consistent with neighbouring severity stages looking alike. With four classes this method forces negative values on average, and the Moderate mean comes from only 2 scans, so only this one reading is reliable.

8.3 Intensity statistics

A random sample of 150 labelled scans showed no NaN or infinite voxels, a minimum of 0 (zeroed background) and a single channel. Typical intensity is tight (p99 about 1.08; mean brain intensity 0.705 with SD 0.012), while the raw maximum varies widely (1.18 to 2.86). The spread therefore lies in the extreme tail and not in the bulk of the intensities, so percentile clipping with a per-volume rescale is sufficient and heavy normalisation is not needed.

8.4 Image quality

Quality metrics (sharpness, SNR, contrast, brain fraction, saturation) flagged 10 of 150 sampled scans using a robust z-score threshold of 3.5. Inspection of the lowest-quality candidates showed that only one (the Moderate scan above) is a true artifact; several others simply have marked atrophy. The class means confirm that these metrics partly track disease:

Class	Sharpness	Contrast	Brain fraction
Nondemented	0.052	0.630	0.266
Very mild	0.074	0.682	0.257
Mild	0.080	0.698	0.262
Moderate	0.128	0.705	0.237

Atrophy widens sulci and ventricles, which raises edge and tissue-to-CSF contrast. Scans should therefore be inspected visually before any are excluded. The saturation metric (about 1e-6, a single voxel) is uninformative.

8.5 Group-mean images

Averaging 15 scans per class (2 for Moderate) shows ventricles and sulci widening progressively from Nondemented to Mild. The Moderate mean and the Nondemented minus Moderate difference map are dominated by the misregistered scan and must not be interpreted.

9. Modelling-Readiness Checks

- **Leakage:** 20 subjects have two sessions, so splitting must be grouped by subject (StratifiedGroupKFold).
- **Stratification:** a 5-fold subject-level split gave 17% to 34% demented per test fold versus 22.9% overall, which is uneven. Repeated cross-validation with several seeds is recommended.
- **Confounds:** age (severe), age-correlated image-quality metrics, and, to a smaller degree, sex and head size (not associated with CDR).

10. Main Finding: Age Confounding

Mean age is 43.8 years for Nondemented subjects (range 18 to 94) versus 76.2, 77.8 and 82.0 for Very mild, Mild and Moderate. Nearly all subjects under 60 are Nondemented, and their CDR of 0 appears to be filled in rather than clinically assessed. A classifier could therefore score well simply by detecting a young brain.

10.1 Age-restricted subset

Restricting to subjects aged 60 and over gives a nearly balanced and age-comparable set:

Label	n	Mean age	SD	Range
Demented (CDR > 0)	100	76.8	7.1	62 to 96
Nondemented	98	75.9	9.0	60 to 94

Demented subjects per decade: 15 (60 to 69), 48 (70 to 79), 37 (80 and over). Nondemented: 25, 35 and 38. The split is slightly uneven within decades (38%, 58% and 49% demented), but it cancels out overall. No subject in this subset has more than one session.

10.2 Baselines

Subject-level 5-fold cross-validation, repeated with 5 seeds, using logistic regression on the 60-plus subset:

Features	Mean AUC
Age only	0.44
nWBV only	0.70
nWBV + eTIV	0.68
nWBV + Age	0.72

Age alone is at chance, confirming that the restricted subset removes the age shortcut (a value slightly below 0.5 is a known cross-validation artifact for a feature with no signal). Brain volume gives a real but modest signal. eTIV adds nothing, consistent with nWBV already being head-size normalised. The seed-to-seed standard deviations (0.004 to 0.010) only reflect fold assignment; with 198 scans the real uncertainty on an AUC near 0.7 is about plus or minus 0.04, so differences between the nWBV-based models are not significant. A nWBV + Age AUC of about 0.72 is the benchmark an image-based model must clearly exceed.

11. Limitations

- The Moderate class has only 2 scans and cannot be modelled; the cause of its atypical scan is unknown (the 2 files with a different naming variant should be checked).
- With 198 usable scans, performance estimates have wide intervals.
- About 70% of the demented subjects are CDR 0.5 (very mild), the hardest stage to separate from normal, so the achievable ceiling is modest.
- MMSE is closely tied to how CDR is assigned and would be label leakage in an image-only model.
- Results apply to adults aged 60 and over, not to dementia versus healthy adults in general.

12. Conclusions and Recommendations

1. Use a binary label (CDR 0 versus CDR greater than 0), or three classes with Mild and Moderate merged. Do not use four classes.
2. Restrict to subjects aged 60 and over (or age-match) to remove the age shortcut. Report age-only and nWBV baselines alongside any model.
3. Split by subject, stratified by label (and age bin), and use repeated cross-validation. Report AUC, sensitivity and specificity, not accuracy alone, with confidence intervals.
4. Images need no resampling. Apply percentile clipping (1st to 99th) and a per-volume rescale. Inspect, do not automatically drop, quality-flagged scans, and review the misregistered Moderate scan.
5. Use only one of eTIV and ASF, and exclude MMSE from image-based models.
6. Start with a tabular baseline and cheap image-derived features (brain volume, CSF fraction), then a pretrained 2D CNN on several axial slices with subject-level folds. A 3D CNN from scratch is likely to overfit on about 200 volumes.

Appendix: Summary Checklist

Item	Result
Sessions / subjects / usable image and CSV pairs	436 / 416 / 436
Class balance (CDR 0 / 0.5 / 1 / 2)	336 / 70 / 28 / 2
Structural missingness	MMSE, Educ, SES absent for most subjects under 60; Delay only for repeat scans
Image uniformity	Shape, voxel size, orientation and dtype uniform (2 files use a different naming variant)
Intensity normalisation needed	Light: percentile clip and per-volume rescale
Split strategy	Subject-level, stratified by label (and age bin), repeated
Confounds noted	Age (severe); age-correlated quality metrics; sex and eTIV not associated with CDR